Stability Optimization in Artificial Agents

gnalens

[2025-10-11 Sat 14:39]

Abstract

This paper develops a formal correspondence between information-theoretic stability (Senn 2025a, 2025b) and the training dynamics of artificial agents. Policy learning in reinforcement learning (RL) and reinforcement learning with human feedback (RLHF) can be interpreted as the maximization of an informational stability reward: the minimization of divergence between successive policy distributions under bounded entropy. When agents couple their internal policies to external reward distributions—such as human feedback—alignment arises as a condition of mutual-information equilibrium. This framework offers a general, mathematically grounded description of alignment as entropy control, rather than as moral or semantic conformity.

1. Introduction

Modern artificial agents are trained to maximize reward functions that capture performance, preference, or human approval (Christiano et al. 2017; Ouyang et al. 2022). These systems can be understood as stochastic processes updating probability distributions over actions in response to feedback. Information geometry (Amari 2016) provides a natural framework for analyzing such processes: every policy update corresponds to a trajectory on the manifold of probability distributions, and learning corresponds to gradient descent on a divergence functional (Kakade 2001; Peters & Schmidhuber 2008).

Building on prior work defining stability as a rate of divergence minimization (Senn 2025a) and its cognitive interpretation via free-energy reduction (Senn 2025b), we show that reinforcement learning implicitly maximizes the same stability reward. We then formalize alignment as an equilibrium condi-

tion in which the agent's policy and the external feedback distribution share maximal mutual information subject to entropy constraints.

2. Background

2.1 Policy Manifold

Let $\pi_\theta(a|s)$ denote a parameterized stochastic policy mapping states $s \in \mathcal{S}$ to actions $a \in \mathcal{A}$ with parameters $\theta \in \Theta$. The space of all such policies forms a manifold $\mathcal{P}_\pi$ equipped with the Fisher–Rao metric

$$g_{ij}(\theta) = \mathbb{E}_{\pi_\theta}[\partial_i \log \pi_\theta(a|s)\, \partial_j \log \pi_\theta(a|s)].$$

Each policy update traces a trajectory π_t on $\mathcal{P}_\pi$.

2.2 KL-Regularized Reinforcement Learning

In entropy-regularized RL (Schulman et al. 2017; Haarnoja et al. 2018), the objective combines expected reward and a penalty on policy divergence:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_t r_t - \alpha\, D_{\mathrm{KL}}(\pi_t || \pi_{t-1}) \right].$$

The KL term constrains the rate of policy change, preventing instability and ensuring smooth adaptation.

2.3 Stability Reward

Following Senn (2025a), define the instantaneous stability reward

$$R_s(t) = -\frac{1}{\delta t}\, D_{\mathrm{KL}}(\pi_{t+\delta} || \pi_t),$$

measured in nats per timestep. Maximizing R_s minimizes the divergence between successive policy distributions, enforcing informational continuity.

3. Policy Stability as Information Optimization

3.1 Gradient Flow

Policy updates can be viewed as a Fisher–natural gradient flow (Amari 1998):

$$\dot{\pi}_t = -\mathrm{grad}_g\, D_{\mathrm{KL}}(\pi_t || \pi_{t-1}),$$

subject to reward-modulated forces proportional to the expected return. The corresponding stability reward satisfies

$$R_s(t) = \langle \dot{\pi}_t, \nabla_\pi \ln \pi_t \rangle_g,$$

and the expected cumulative reward

$$\mathbb{E}\left[\sum_t R_s(t)\right]$$

serves as a regularizer ensuring convergence toward a stable optimum.

3.2 Equivalence to Free-Energy Descent

Let the policy induce a distribution over trajectories $\tau = (s_1, a_1, \dots)$ with probability $p_\pi(\tau)$. The negative log-evidence $-\ln p_\pi(\tau)$ corresponds to cumulative prediction error. Minimizing the divergence between consecutive $p_\pi(\tau)$ is equivalent to minimizing the temporal derivative of variational free energy (Friston 2010). Thus, reinforcement learning realizes the same informational dynamic as cognitive free-energy minimization, differing only in the definition of reward.

4. Mutual-Information Coupling and Alignment

4.1 External Reward Distribution

Let $p_H(a|s)$ denote a human-generated or externally defined preference distribution. During RLHF training, the agent's policy $\pi_\theta(a|s)$ receives scalar feedback approximating $\log p_H(a|s)$ (Christiano et al. 2017; Ouyang et al. 2022). Define coupling strength κ via mutual information:

$$I(\pi; H) = \sum_{a,s} p(a,s)\, \log \frac{\pi_\theta(a|s)}{p_H(a|s)}.$$

4.2 Alignment Equilibrium

The joint divergence between agent and feedback distributions is

$$D_{\text{joint}} = D_{\text{KL}}(\pi || p_H) + D_{\text{KL}}(p_H || \pi).$$

Gradient descent on D_{joint} under symmetric coupling κ yields

$$\partial_t I(\pi; H) \geq 0,$$

implying monotonic increase of mutual information until equilibrium $\nabla_\pi D_{\text{joint}} = 0$. At this point, the agent's policy achieves informational alignment with the external distribution—an equilibrium of stability rather than of meaning.

5. Regularization and Overstability

Over-optimization of stability leads to pathologies analogous to mode collapse in generative modeling or reward hacking in RL (Amodei et al. 2016). When $\partial_t D_{\text{KL}}(\pi_{t+\delta}||\pi_t) \to 0$ too rapidly, exploration ceases, and the system converges prematurely to low-entropy attractors. Conversely, insufficient regularization allows divergence explosion, producing instability or catastrophic forgetting. Optimal training therefore requires balancing the stability reward with entropy production:

$$\mathcal{L} = -R_s + \beta\, H(\pi_t),$$

where β controls the exploration–stability trade-off.

6. Discussion

1. **Alignment as entropy control.** Alignment emerges when mutual information between the agent's policy and the feedback source saturates under bounded entropy production. This reframes value alignment as an informational, not moral, problem.

2. **Unified learning geometry.** Gradient-based learning in both biological and artificial agents implements the same divergence-minimization principle on the manifold of probability distributions (Amari 1998; Senn 2025b).

3. **Design implications.** Multi-objective or plural feedback sources can be modeled as multiple coupled distributions $\{p_{H_i}\}$, each exerting partial information pressure. Anti-stable agents—those designed to sustain controlled divergence—may support creative or exploratory behaviors.

7. Conclusion

Artificial agents trained by reinforcement or feedback learn by minimizing divergence between successive policy states. This process, when formalized

through information geometry, constitutes the maximization of informational stability. Alignment with human preferences arises when the agent and feedback distributions reach a mutual-information equilibrium. The framework situates alignment, exploration, and robustness within a unified geometric theory of stability optimization.

References

- Amari, S. (1998). "Natural Gradient Works Efficiently in Learning." **Neural Computation**, 10(2), 251–276.
- Amari, S. (2016). **Information Geometry and Its Applications**. Springer.
- Amodei, D., et al. (2016). "Concrete Problems in AI Safety." **arXiv:1606.06565**.
- Christiano, P. F., Leike, J., Brown, T., et al. (2017). "Deep Reinforcement Learning from Human Preferences." **Advances in Neural Information Processing Systems**, 30.
- Cover, T., & Thomas, J. (2006). **Elements of Information Theory** (2nd ed.). Wiley.
- Friston, K. J. (2010). "The Free-Energy Principle: A Unified Brain Theory?" **Nature Reviews Neuroscience**, 11(2), 127–138.
- Haarnoja, T., Zhou, A., Abbeel, P., & Levine, S. (2018). "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor." **ICML 2018.**
- Kakade, S. M. (2001). **A Natural Policy Gradient.** MIT.
- Ouyang, L., Wu, J., Jiang, X., et al. (2022). "Training Language Models to Follow Instructions with Human Feedback." **arXiv:2203.02155.**
- Peters, J., & Schmidhuber, J. (2008). "Natural Actor-Critic." **Neurocomputing**, 71(7-9), 1180–1190.
- Schulman, J., Chen, X., & Abbeel, P. (2017). "Proximal Policy Optimization Algorithms." **arXiv:1707.06347.**
- Senn, E. (2025a). **Information-Theoretic Stability as a Reward Function.** arXiv:math.IT.

- Senn, E. (2025b). **Stability Dynamics in Cognitive Systems: A Predictive-Processing Interpretation.** arXiv:cs.AI.